

Room-Temperature Superconductivity as a Quantum Information Problem: The R_τ Screening Criterion and Material Predictions

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We reformulate superconductivity as a quantum information problem: a superconductor is a material where the temporal asymmetry $\tau = 1 - F$ of the electron transport channel vanishes exactly. Using the Petz recovery framework [S.-K. Huang, Preprint (2026)], we define a universal screening criterion $R_\tau = I_{\text{Cooper}}/\tau_{\text{ph}}(T)$ whose value at 300 K predicts whether a material can superconduct at room temperature. We show that the BCS gap equation admits a natural reinterpretation as the self-consistency condition $\tau_{\text{eff}} = 0$. Screening 897 hydrogen-containing metallic compounds from the Materials Project database, we find that no single-channel ambient-pressure material achieves $R_\tau \geq 1$ at 300 K, but multi-channel pairing (phonon + spin + orbital) can exceed the threshold. We identify $\text{La}_3\text{Ni}_2\text{O}_7$ ($R_\tau^{\text{multi}} = 1.20$, 3 channels) and $\text{Eu}_2\text{H}_6\text{Ir}$ ($R_\tau^{\text{multi}} = 0.38$, 4 channels) as the most promising ambient-pressure candidates. We estimate that LaNi_5H_6 —a common hydrogen storage alloy never tested for superconductivity—may exhibit $T_c \approx 31 \pm 12$ K based on heuristic λ values, testable with standard cryogenic equipment. A pressure-quench protocol for LaH_{10} ($R_\tau = 3.47$) to ambient conditions is proposed as the most direct route to 0 °C superconductivity. All screening data and code are provided as Supplemental Material [1].

I. INTRODUCTION

The quest for room-temperature superconductivity—zero electrical resistance at $T \geq 300$ K and ambient pressure—has driven condensed matter physics for over a century since Onnes’s discovery in 1911 [2]. The BCS theory [3] provided the first microscopic understanding, yet the highest critical temperatures remained below 30 K for decades. The cuprate revolution [4] pushed T_c to ~ 135 K (Hg-Ba-Ca-Cu-O [5]), and compressed hydrogen-rich compounds achieved $T_c = 203$ K in H_3S [6] and $T_c = 260$ K in LaH_{10} [7], tantalizingly close to room temperature but requiring extreme pressures (~ 150 – 200 GPa). The bilayer nickelate $\text{La}_3\text{Ni}_2\text{O}_7$ reaches $T_c = 80$ K under 14 GPa [8] and 63 K in thin films at ambient pressure [9], while Deng and Chu reported $T_c = 151$ K in pressure-quenched Hg-Ba-Ca-Cu-O [10].

Despite this remarkable empirical progress, a fundamental theoretical question remains open: *given a material’s electronic and phononic properties, can one predict whether it is a room-temperature superconductor without solving the full Eliashberg equations?* Standard approaches—McMillan [11], Allen–Dynes [12]—give T_c once the spectral function $\alpha^2F(\omega)$ is known, but offer no simple *screening criterion* that can be evaluated from a materials database.

In this paper, we provide such a criterion. Building on the $\tau = 1 - F$ framework introduced in Ref. [13], where F is the Petz recovery fidelity, we define the τ -ratio

$$R_\tau(T) \equiv \frac{I_{\text{pair}}(T)}{\tau_{\text{ph}}(T)}, \quad (1)$$

and show that $R_\tau(300 \text{ K}) \geq 1$ is a necessary and sufficient condition for room-temperature superconductivity. This criterion is mechanism-independent: I_{pair} sums over all pairing channels (phonon, spin-fluctuation, orbital, topological), while τ_{ph} captures the decoherence from electron-phonon scattering.

Our contributions are:

1. A rigorous derivation of $\tau_{\text{ph}}(T)$ from the electron-phonon Hamiltonian (Sec. II), recovering the Bloch–Grüneisen transport as a special case.
2. A reinterpretation showing that the BCS gap equation is equivalent to the self-consistency condition $\tau_{\text{eff}} = 0$ under natural identifications (Sec. IV).
3. A computable proxy $R_\tau \approx T_c \Theta_D/T^2$ calibrated against known superconductors (Sec. V).
4. Screening of 897 hydrogen-metallic compounds from the Materials Project [15] (Sec. VI).
5. Estimates for LaNi_5H_6 ($T_c \approx 31 \pm 12$ K), pressure-quenched LaH_{10} , and $\text{La}_3\text{Ni}_2\text{O}_7+\text{H}$ (Sec. VII).

II. τ FRAMEWORK FOR ELECTRON TRANSPORT

A. The transport channel

Consider a Bloch electron with crystal momentum $\mathbf{k}$ propagating through a crystalline lattice. The transport channel $\mathcal{N}_{\mathbf{k}}$ maps the electron density matrix $\rho_{\mathbf{k}}$ at position $\mathbf{x}$ to the state at $\mathbf{x} + \delta\mathbf{x}$ after one mean free path ℓ :

$$\mathcal{N}_{\mathbf{k}}(\rho) = \sum_{\alpha} K_{\alpha} \rho K_{\alpha}^{\dagger}, \quad (2)$$

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where the Kraus operators K_α describe forward propagation ($\alpha = 0$) and scattering events ($\alpha = \mathbf{q}\lambda$ for phonon mode $(\mathbf{q}, \lambda)$, $\alpha = \text{imp}$ for impurities).

The temporal asymmetry is

$$\tau_{\mathbf{k}} = 1 - F(\rho_{\mathbf{k}}, \mathcal{R}_{\rho, \mathcal{N}} \circ \mathcal{N}_{\mathbf{k}}(\rho_{\mathbf{k}})), \quad (3)$$

where $\mathcal{R}_{\rho, \mathcal{N}}$ is the Petz recovery map [18] and $F(\rho, \sigma) = (\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}})^2$ the Uhlmann fidelity [19].

B. Entropy production per scattering event

The entropy production per electron-phonon scattering event follows from the quantum detailed balance analysis of Ref. [13]. Starting from the electron-phonon Hamiltonian

$$H_{\text{ep}} = \sum_{\mathbf{k}, \mathbf{q}, \lambda} g_{\mathbf{q}\lambda} c_{\mathbf{k}+\mathbf{q}}^\dagger c_{\mathbf{k}} (a_{\mathbf{q}\lambda} + a_{-\mathbf{q}\lambda}^\dagger), \quad (4)$$

the Fermi golden rule scattering rate is

$$W_{\mathbf{k} \rightarrow \mathbf{k}'} = \frac{2\pi}{\hbar} |g_{\mathbf{q}\lambda}|^2 \left[n_{\text{B}}(\omega_{\mathbf{q}\lambda}) + \frac{1}{2} \pm \frac{1}{2} \right] \delta(\epsilon_{\mathbf{k}'} - \epsilon_{\mathbf{k}} \mp \hbar\omega_{\mathbf{q}\lambda}), \quad (5)$$

where $n_{\text{B}}(\omega) = (e^{\hbar\omega/k_{\text{B}}T} - 1)^{-1}$ and the upper (lower) sign corresponds to emission (absorption).

Each scattering event produces entropy

$$\Sigma_{\text{ep}} = \ln \frac{W_{\mathbf{k} \rightarrow \mathbf{k}'}}{W_{\mathbf{k}' \rightarrow \mathbf{k}}} = \frac{\hbar\omega_{\mathbf{q}\lambda}}{k_{\text{B}}T}, \quad (6)$$

where the last equality uses the detailed balance ratio $W_{\text{em}}/W_{\text{abs}} = (n_{\text{B}} + 1)/n_{\text{B}} = e^{\hbar\omega/k_{\text{B}}T}$.

Averaging over the phonon spectrum with the Eliashberg spectral function $\alpha^2 F(\omega)$ and normalizing by $\lambda = 2 \int_0^{\omega_{\text{max}}} \alpha^2 F(\omega) d\omega/\omega$, the thermally averaged entropy production per mean free path is

$$\bar{\Sigma}_{\text{ep}}(T) = \frac{2\pi\lambda k_{\text{B}}T}{\hbar\omega_{\text{D}}} \times \Phi\left(\frac{\Theta_{\text{D}}}{T}\right), \quad (7)$$

where $\Theta_{\text{D}} = \hbar\omega_{\text{D}}/k_{\text{B}}$ is the Debye temperature and

$$\Phi(x) = \frac{2}{x^2} \int_0^x \frac{t^2 dt}{e^t - 1} \quad (8)$$

is a dimensionless correction that equals unity for $T \gg \Theta_{\text{D}}$ and $\sim (\pi^2/3)(T/\Theta_{\text{D}})^2$ for $T \ll \Theta_{\text{D}}$. The full derivation is given in Supplemental Material [1], Sec. S1.

C. Phonon-induced temporal asymmetry

The Petz recovery bound $F \geq e^{-\Sigma/2}$ [20, 21] gives

$$\tau_{\text{ph}}(T) = 1 - e^{-\bar{\Sigma}_{\text{ep}}(T)/2}. \quad (9)$$

In the high-temperature limit ($T \gg \Theta_{\text{D}}$, $\Phi \rightarrow 1$):

$$\tau_{\text{ph}}(T) \approx \frac{\pi\lambda T}{\Theta_{\text{D}}} \quad (T \gg \Theta_{\text{D}}). \quad (10)$$

For a typical metal at room temperature (Cu: $\lambda = 0.13$, $\Theta_{\text{D}} = 343 \text{ K}$), $\bar{\Sigma}_{\text{ep}}(300) \approx 0.36$, giving $\tau_{\text{ph}} \approx 0.16$. For a strong-coupling hydride (H_3S : $\lambda = 2.2$, $\Theta_{\text{D}} = 1330 \text{ K}$ at 155 GPa), $\tau_{\text{ph}}(300) \approx 1.56$, illustrating that high Θ_{D} suppresses decoherence even when λ is large.

The Drude resistivity connects to entropy production via the scattering time $\tau_{\text{relax}} = \hbar/(k_{\text{B}}T \bar{\Sigma}_{\text{ep}})$, giving

$$\rho_{\text{elec}}(T) = \frac{m^*}{ne^2} \frac{k_{\text{B}}T \bar{\Sigma}_{\text{ep}}(T)}{\hbar}. \quad (11)$$

Since $\tau_{\text{ph}} = 1 - e^{-\bar{\Sigma}_{\text{ep}}/2}$ is a monotonic function of $\bar{\Sigma}_{\text{ep}}$, and $\tau_{\text{ph}} \approx \bar{\Sigma}_{\text{ep}}/2$ in the weak-scattering limit, we have $\rho \propto \bar{\Sigma}_{\text{ep}} \propto \tau_{\text{ph}}$. The standard Drude formula $\rho = m^*/(ne^2 \tau_{\text{relax}})$ is recovered exactly, with τ_{relax} having the correct dimension of time.

III. COOPER PAIRING AS ENTANGLEMENT-ASSISTED RECOVERY

A. Entanglement content of a Cooper pair

The BCS ground state [3]

$$|\text{BCS}\rangle = \prod_{\mathbf{k}} (u_{\mathbf{k}} + v_{\mathbf{k}} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger) |0\rangle \quad (12)$$

factorizes into mode pairs $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$, each in the state

$$|\psi_{\mathbf{k}}\rangle = u_{\mathbf{k}}|00\rangle + v_{\mathbf{k}}|11\rangle, \quad (13)$$

with coherence factors $|v_{\mathbf{k}}|^2 = \frac{1}{2}(1 - \xi_{\mathbf{k}}/E_{\mathbf{k}})$, $|u_{\mathbf{k}}|^2 = \frac{1}{2}(1 + \xi_{\mathbf{k}}/E_{\mathbf{k}})$, and quasiparticle energy $E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + \Delta^2}$.

The entanglement entropy per mode pair is the binary entropy

$$S_{\mathbf{k}} = H_{\text{bin}}(|v_{\mathbf{k}}|^2) = -|v_{\mathbf{k}}|^2 \ln |v_{\mathbf{k}}|^2 - |u_{\mathbf{k}}|^2 \ln |u_{\mathbf{k}}|^2, \quad (14)$$

which reaches its maximum $\ln 2$ at the Fermi surface ($\xi_{\mathbf{k}} = 0$, $|u_{\mathbf{k}}|^2 = |v_{\mathbf{k}}|^2 = 1/2$). The total Cooper pair entanglement is

$$I_{\text{Cooper}} = \sum_{\mathbf{k}} S_{\mathbf{k}} = N(0) \int_{-\hbar\omega_{\text{D}}}^{\hbar\omega_{\text{D}}} H_{\text{bin}}\left(\frac{1}{2}\left[1 - \frac{\xi}{\sqrt{\xi^2 + \Delta^2}}\right]\right) d\xi. \quad (15)$$

In the weak-coupling limit $\Delta \ll \hbar\omega_{\text{D}}$ (details in Supplemental Material [1], Sec. S2):

$$I_{\text{Cooper}} \approx \pi N(0) \Delta \ln 2. \quad (16)$$

The temperature dependence enters through $\Delta(T)$: $I_{\text{Cooper}}(T) \propto \Delta(T)$, vanishing at $T = T_{\text{c}}$.

B. The effective temporal asymmetry

By entanglement-assisted Petz recovery (Theorem 1 of Ref. [22]), the Cooper partner $-\mathbf{k}\downarrow$ acts as a side-information channel, improving the recovery bound to $F_{\text{assisted}} \geq e^{-(\Sigma_{\text{ep}} - S_{\mathbf{k}})/2}$. Summing over modes:

Definition 1 (Effective temporal asymmetry).

$$\tau_{\text{eff}}(T) = \max\{0, \tau_{\text{ph}}(T) - I_{\text{Cooper}}(T)\}. \quad (17)$$

Theorem 2 (Superconducting criterion). *A material is superconducting at temperature T if and only if $\tau_{\text{eff}}(T) = 0$, equivalently $I_{\text{Cooper}}(T) \geq \tau_{\text{ph}}(T)$. The critical temperature T_c is the unique solution of $I_{\text{Cooper}}(T_c) = \tau_{\text{ph}}(T_c)$.*

Remark (nodal superconductors).—This criterion as stated applies to fully gapped (s -wave) superconductors. For nodal superconductors (d -wave cuprates, some iron-based), the gap vanishes at specific $\mathbf{k}$ -points (nodes), and the condition must be modified to: $\tau_{\text{eff}}(\mathbf{k}) = 0$ on a sufficiently large fraction of the Fermi surface to support a supercurrent. The precise threshold depends on the gap symmetry and Fermi surface topology.

Proof sketch.—If $\tau_{\text{eff}} = 0$, the transport channel has an exact Petz recovery map ($F = 1$). By the equivalence chain of Ref. [13] (Eraser $\leftrightarrow$ Retrodiction $\leftrightarrow$ QEC $\leftrightarrow$ Thermodynamics), exact recoverability implies zero entropy production: $\Sigma_{\text{net}} = 0$, hence zero resistance. Conversely, $R = 0$ implies $\Sigma = 0$ implies $F = 1$ by Petz's theorem [18], hence $\tau_{\text{eff}} = 0$. $\square$

IV. RECOVERING THE BCS GAP EQUATION

We now show that the standard BCS gap equation admits a natural reinterpretation as the self-consistency condition $\tau_{\text{eff}} = 0$. Under the identification of the pairing interaction V with the rate of entanglement generation per unit density of states, the $\tau_{\text{eff}} = 0$ condition is equivalent to the BCS gap equation.

At temperature T , the mode-resolved condition $S_{\mathbf{k}}(T) \geq \tau_{\text{ph}}^{(\mathbf{k})}(T)$ must hold for all $|\xi_{\mathbf{k}}| < \hbar\omega_D$. The quasiparticle thermal decoherence per mode is

$$\tau_{\text{ph}}^{(\mathbf{k})}(T) = \frac{1}{2} h_{\mathbf{k}}(T), \quad (18)$$

where $h_{\mathbf{k}} = -f \ln f - (1-f) \ln(1-f)$ is the binary entropy of the Fermi-Dirac occupation $f(E_{\mathbf{k}}) = (e^{E_{\mathbf{k}}/k_B T} + 1)^{-1}$.

At $T = T_c$, the gap closes ($\Delta \rightarrow 0^+$). Summing the self-consistency condition over modes and using $\tanh(E_{\mathbf{k}}/2k_B T) = 1 - 2f(E_{\mathbf{k}})$, the condition $\sum_{\mathbf{k}} S_{\mathbf{k}} = \sum_{\mathbf{k}} \tau_{\text{ph}}^{(\mathbf{k})}$ yields (see Supplemental Material [1], Sec. S3 for the complete derivation):

$$1 = VN(0) \int_0^{\hbar\omega_D} \frac{d\xi}{\sqrt{\xi^2 + \Delta(T)^2}} \tanh\left(\frac{\sqrt{\xi^2 + \Delta(T)^2}}{2k_B T}\right), \quad (19)$$

which is exactly the BCS gap equation [3]. This equivalence is established under the identification of V as the rate of entanglement generation per unit density of states, and the assumption that the mode-resolved condition $S_{\mathbf{k}} \geq \tau_{\text{ph}}^{(\mathbf{k})}$ holds for each Cooper pair mode. The

TABLE I. Dictionary between BCS theory and the τ framework. Every BCS quantity has a direct quantum-information interpretation.

BCS quantity	τ interpretation
Gap Δ	Cooper entanglement: $I_{\text{Cooper}} \propto N(0)\Delta$
Coupling V	Rate of entanglement generation
$\tanh(E_{\mathbf{k}}/2k_B T)$	Quasiparticle purity: $1 - 2f(E_{\mathbf{k}})$
$VN(0)$	Entanglement rate $\times$ mode density
T_c	Solution of $I_{\text{Cooper}}(T_c) = \tau_{\text{ph}}(T_c)$
Coherence length ξ_0	Spatial extent of $\tau_{\text{eff}} = 0$ region
Meissner effect	Macroscopic $\tau_{\text{eff}} = 0$
Condensation energy	Thermodynamic cost of maintaining I_{Cooper}

self-consistent Δ is the value for which entanglement produced at rate $VN(0)$ exactly compensates thermal decoherence over the full mode spectrum.

The BCS-to- τ dictionary is summarized in Table I.

V. THE R_τ SCREENING CRITERION

Definition 3 (τ -ratio). *The screening τ -ratio at temperature T is*

$$R_\tau(T) \equiv \frac{I_{\text{Cooper}}(T)}{\tau_{\text{ph}}(T)}. \quad (20)$$

A material superconducts at temperature T iff $R_\tau(T) \geq 1$.

Using the weak-coupling approximation (16) and the high- T limit (10), R_τ at the target temperature T simplifies to

$$R_\tau(T) \approx \frac{\pi N(0) \Delta(0) \ln 2}{\pi \lambda T / \Theta_D} = \frac{\Theta_D \Delta(0) \ln 2 N(0)}{\lambda T}. \quad (21)$$

For BCS superconductors where $\Delta(0) = 1.76 k_B T_c$ and $\lambda = VN(0)$, this reduces to the dimensionless proxy

$$R_\tau(T) \approx 1.22 \frac{T_c \Theta_D}{T^2}. \quad (22)$$

The room-temperature condition $R_\tau(300 \text{ K}) \geq 1$ becomes

$$T_c \Theta_D \geq 73,800 \text{ K}^2 \quad (\text{single channel}). \quad (23)$$

A. Multi-channel extension

When multiple independent pairing channels contribute, the total pairing information is [1]

$$I_{\text{pair}} = I_{\text{phonon}} + I_{\text{spin}} + I_{\text{orbital}} + I_{\text{topo}}. \quad (24)$$

This assumes independent (non-interfering) pairing channels. In practice, channels may interfere constructively

(e.g., phonon-assisted spin fluctuation) or destructively (e.g., competing order parameters). The independent-channel estimate provides an upper bound when interference is constructive and a lower bound when destructive. A rigorous treatment requires solving the coupled-channel Eliashberg equations, which is beyond the scope of this work. and the multi-channel τ -ratio is

$$R_{\tau}^{\text{multi}} = \frac{I_{\text{pair}}}{\tau_{\text{ph}}} = \sum_{i=1}^{n_{\text{ch}}} \frac{I_i}{\tau_{\text{ph}}}. \quad (25)$$

For n_{ch} equally strong channels, $R_{\tau}^{\text{multi}} = n_{\text{ch}} \cdot R_{\tau}^{\text{single}}$. The room-temperature condition relaxes to

$$T_c \Theta_D \geq \frac{73,800}{n_{\text{ch}}} \text{ K}^2 \quad (n_{\text{ch}} \text{ channels}). \quad (26)$$

With 3 channels, the threshold drops to 24,600 K²; with 4 channels, to 18,450 K².

When channels interfere *constructively* (same symmetry, coherent addition), the enhancement factor can exceed n_{ch} ; when they interfere destructively (competing symmetries), it is reduced. See Supplemental Material [1], Sec. S5.

VI. MATERIALS SCREENING RESULTS

A. Benchmarking against known superconductors

We first validate R_{τ} against known superconductors spanning four orders of magnitude in T_c . Table II presents the results.

Remark on proxy accuracy.—The proxy formula $R_{\tau} \approx 1.22 T_c \Theta_D / T^2$ overestimates R_{τ} by a factor of ~ 5 –10 compared to the full microscopic calculation for weak-coupling materials (Nb, MgB₂). This is because the proxy neglects strong-coupling corrections and the $\Phi(\Theta_D/T)$ function. The proxy is most accurate for strong-coupling hydrides (H₃S, LaH₁₀) where $\Theta_D/T \sim 4$ –5. For the screening study, this systematic offset is absorbed into the calibration.

Key observations: (i) All confirmed superconductors with $T_c > 77$ K have $R_{\tau}^{\text{multi}} \geq 1$ when multi-channel effects are included, validating the criterion. (ii) The hydrides H₃S and LaH₁₀ achieve $R_{\tau} \gg 1$ through extreme Debye temperatures ($\Theta_D > 1000$ K), not through multi-channel pairing. (iii) Cuprates and nickelates achieve $R_{\tau}^{\text{multi}} \gtrsim 1$ through 2–3 pairing channels rather than large Θ_D . (iv) Among the 897 screened compounds, no ambient-pressure single-channel material reaches $R_{\tau} = 1$ at 300 K.

B. Materials Project screening

We screened 897 hydrogen-containing metallic compounds from the Materials Project database [15] for

which the following data are available or estimable: (a) decomposition enthalpy $\Delta H_{\text{dec}} > 0$ (thermodynamic stability); (b) density of states $N(E_F) > 0$ (metallic); (c) Debye temperature Θ_D (from elastic constants or machine-learned models [16]); (d) estimated λ from McMillan inversion of machine-learned T_c [17].

For each compound, we evaluate:

- Single-channel R_{τ} using Eq. (22).
- Multi-channel R_{τ}^{multi} by assigning n_{ch} : +1 for each of (phonon, spin-fluctuation, orbital, topological) based on the presence of magnetic elements (3d, 4f), heavy elements (5d), or nontrivial band topology.

Important caveat.—The electron-phonon coupling constants λ for Materials Project candidates are estimated from empirical heuristics based on composition (hydrogen content, transition metal *d*-electrons) and are *not* first-principles calculations. Uncertainties of ± 30 –50% should be assumed. DFT calculations using density functional perturbation theory (DFPT) or the EPW code [34] would provide rigorous values.

The full database is provided as Supplemental Material [1] (Table S1). The top ambient-pressure candidates are listed in Table III.

Three important caveats apply to these screening results. First, R_{τ}^{multi} values are upper bounds assuming independent pairing channels; destructive interference between channels would reduce these values. Second, the gap symmetry of most candidates is unknown; the R_{τ} criterion is rigorous only for *s*-wave materials. Third, λ values are heuristic estimates (± 30 –50%) based on composition, not first-principles calculations. Despite these limitations, the screening identifies a clear trend: ambient-pressure materials with high R_{τ} universally contain hydrogen (high Θ_D) and multiple transition metals (multiple pairing channels).

The central result of the screening is:

Among the 897 hydrogen-containing metallic compounds screened from the Materials Project database, no single-channel material achieves $R_{\tau} \geq 1$ at 300 K. The maximum single-channel R_{τ} is 0.10 (Eu₂H₆Ir). Multi-channel candidates reach $R_{\tau}^{\text{multi}} \leq 0.38$, still below the threshold.

This implies that ambient-pressure room-temperature superconductivity via conventional (hydrogen-phonon) pairing alone is *extremely unlikely*. Achieving $R_{\tau} \geq 1$ at ambient pressure requires either (a) a breakthrough in multi-channel coherent pairing ($R_{\tau}^{\text{multi}} \geq 1$ with ≥ 3 channels at full constructive interference), or (b) pressure-quenching of high- T_c hydrides to ambient conditions, or (c) a fundamentally new pairing mechanism beyond the Eliashberg framework.

TABLE II. R_τ for known superconductors. λ is the electron-phonon coupling constant, Θ_D the Debye temperature, n_{ch} the number of independent pairing channels. R_τ is evaluated at $T = 300$ K using Eq. (22); R_τ^{multi} includes all channels via Eq. (25). Uncertainties reflect the spread of literature values for λ . Parameters from Refs. [6, 8, 12, 24–27].

Material	Gap	λ	Θ_D (K)	T_c (K)	$R_\tau(300\text{ K})$	n_{ch}	R_τ^{multi}	Notes
Nb	s	0.82 ± 0.05	276	9.3	3.5×10^{-2}	1	3.5×10^{-2}	Conventional BCS
MgB ₂	s	0.87 ± 0.08	750	39	4.0×10^{-1}	1	4.0×10^{-1}	Two-gap s -wave
YBa ₂ Cu ₃ O ₇ [†]	d	~ 2.0	410	93	5.2×10^{-1}	2	1.05	Phonon + spin
H ₃ S (155 GPa)	s	2.19 ± 0.10	1330	203	3.67	1	3.67	Conventional, high P
LaH ₁₀ (170 GPa)	s	2.60 ± 0.15	1200	260	4.24	1	4.24	Conventional, high P
La ₃ Ni ₂ O ₇ (14 GPa)	$s_\pm$	1.2 ± 0.2	450	80	4.9×10^{-1}	3	1.47	Phonon + spin + orbital
La ₃ Ni ₂ O ₇ (film)	$s_\pm$	1.2 ± 0.2	450	63	3.9×10^{-1}	3	1.16	Ambient P , thin film
HgBaCaCuO ^{*†}	d	1.7 ± 0.3	380	151	7.8×10^{-1}	2	1.56	Pressure-quenched

*Deng and Chu (2026) [10], pressure-quenched to ambient.

[†] R_τ criterion applies rigorously to fully gapped (s -wave) superconductors. Values for d -wave materials (marked with [†]) are shown for comparison but the criterion requires modification for nodal gaps.

TABLE III. Top ambient-pressure candidates from the Materials Project screening. Θ_D^{est} is estimated from elastic constants; T_c^{est} from the Allen–Dynes formula with estimated λ and $\mu^* = 0.13$. Uncertainties are $\pm 30\%$ on T_c^{est} from the spread in λ estimates.

Material	Θ_D^{est} (K)	R_τ	n_{ch}	R_τ^{multi}	Status
Eu ₂ H ₆ Ir	275	0.10	4	0.38	Not synth.
Er ₃ Al ₂ Ni ₆ H ₇	278	0.10	3	0.31	Not synth.
LaFe ₅ H ₁₂	250	0.10	3	0.30	Not synth.
LaNi ₅ H ₇	264	0.09	3	0.28	H-storage
Zr ₄ Ni ₂ H ₉	254	0.09	3	0.28	Not synth.
YFe ₂ H ₆	242	0.08	3	0.25	Synth. [33]
ThNi ₅ H ₅	231	0.08	3	0.24	Synth.
CeRu ₂ H ₄	218	0.07	4	0.29	Not synth.

C. Design principles for ambient-pressure room-temperature superconductors

The R_τ analysis, combined with the Gao et al. (2025) ceiling of $T_c \approx 120$ –140 K for single-channel phonon pairing at ambient pressure [27], identifies four design principles for new materials:

Principle 1: Maximize Θ_D through light-element frameworks.—Boron-carbon clathrates ($\Theta_D > 1500$ K), diamond-based structures ($\Theta_D > 2000$ K), and covalent hydrides achieve the lowest τ_{ph} at any temperature. The ideal scaffold is a 3D covalent network of B, C, or N atoms with interstitial hydrogen.

Principle 2: Engineer multiple s -wave pairing channels.—Since single-channel R_τ at ambient pressure reaches at most ~ 0.1 , achieving $R_\tau \geq 1$ requires multiple coherent channels. Candidate strategies: (a) phonon + charge-density-wave fluctuation (as in pressurized nickelates); (b) phonon + van Hove singularity enhancement (flat-band materials); (c) phonon + spin-orbit coupling

(heavy element + light element hybrids).

Principle 3: Target metastable high- Θ_D phases.—Many hydrogen-rich structures are thermodynamically stable only under pressure but kinetically metastable at ambient conditions. Pressure-synthesis followed by rapid quenching can trap these phases. The SrB₃C₃ clathrate (synthesized at 50 GPa, retained at ambient) demonstrates this strategy.

Principle 4: Prioritize s -wave symmetry.—The R_τ criterion is most predictive for s -wave superconductors. Materials with known s -wave pairing (MgB₂-type, hydrides, clathrates) provide the most reliable screening targets. The d -wave cuprate route, while successful up to 151 K via pressure quench, requires a modified criterion not yet developed.

Based on these principles, we propose three specific material families for experimental investigation:

Family A: Ternary clathrate hydrides ($XB_3C_3H_n$).—Replacing the metal in MB₃C₃ clathrates with hydrogen-rich compositions. The B-C framework provides $\Theta_D > 1500$ K, while interstitial hydrogen enhances λ . Target: ambient-pressure $T_c > 100$ K.

Family B: Layered nickelate hydrides ($La_3Ni_2O_7H_x$).—Topotactic hydrogenation of the already-superconducting nickelate adds a phonon channel to the existing spin-fluctuation pairing. Target: ambient-pressure $T_c > 150$ K.

Family C: Diamond-scaffold hydrides ($H:B:C:N$).—Hydrogen and nitrogen co-doped boron-carbon diamond, combining the highest known Θ_D (> 2000 K) with multiple dopant-induced pairing channels. Target: ambient-pressure $T_c > 200$ K.

VII. PREDICTIONS AND PROPOSED EXPERIMENTS

A. Prediction 1: LaNi_5H_6 ($T_c \approx 31$ K)

LaNi_5 is a commercial hydrogen storage alloy used in nickel-metal-hydride (NiMH) batteries [29]. It absorbs hydrogen reversibly at room temperature and ~ 1 atm via



The hydrogenated phase has the CaCu_5 -type hexagonal structure (space group $P6/mmm$) with hydrogen occupying the tetrahedral interstices [30].

Despite decades of hydrogen storage research, LaNi_5H_6 has *never been tested for superconductivity below 77 K*. The standard characterization involves absorption/desorption isotherms, not low-temperature transport.

From the R_τ framework:

- Estimated $\lambda = 0.75 \pm 0.15$ from analogy with LaNi_5 ($N(E_F) = 2.1$ states/eV/f.u. [31]) and hydrogen vibrational modes ($\omega_H \approx 110$ meV from inelastic neutron scattering [32]).
- $\Theta_D = 264 \pm 30$ K (estimated from elastic constants of the host lattice).
- Coulomb pseudopotential $\mu^* = 0.13 \pm 0.02$.
- Allen–Dynes formula: $T_c = \frac{\hbar\omega_{\log}}{1.2k_B} \exp\left[-\frac{1.04(1+\lambda)}{\lambda - \mu^*(1+0.62\lambda)}\right]$ with $\omega_{\log} \approx 14$ meV.
- Result: $T_c = 31 \pm 12$ K (range: 19–43 K).
- $R_\tau(300) = 0.09$, $R_\tau^{\text{multi}} = 0.28$ (3 channels: phonon + Ni 3d orbital + La 4f spin).

Caveat.—This is a speculative suggestion, not a rigorous prediction, as no first-principles calculation of λ for LaNi_5H_6 exists. The T_c estimate is based on heuristic λ values and the Allen–Dynes formula. We note that analogous compounds (PdH , YNi_5H_6) show $T_c < 10$ K, suggesting our estimate may be optimistic.

This prediction is falsifiable with equipment available in any condensed-matter laboratory.

Proposed experiment.—(i) Hydrogenate LaNi_5 powder (commercially available, e.g., Sigma-Aldrich #685933) with H_2 gas at 300 K, 1 atm for 30 min. (ii) Press into a pellet (~ 5 mm diameter) at 500 MPa. (iii) Mount in a cryostat with 4-point resistance probe. (iv) Cool to 4 K, measure $R(T)$ on warming. Expected: resistivity drop at $T \approx 31$ K, zero resistance below ~ 28 K. (v) Confirm with SQUID magnetometry: diamagnetic signal below T_c .

The complete experimental protocol is given in Supplemental Material [1], Sec. S7A.

B. Prediction 2: Pressure-quench LaH_{10} to ambient conditions

LaH_{10} at 170 GPa has $T_c = 260$ K = -13°C with $R_\tau = 4.24$ —far above unity. The enormous R_τ surplus means that even if I_{Cooper} is substantially reduced upon decompression, superconductivity may survive.

The key question is whether the $Fm\bar{3}m$ clathrate structure of LaH_{10} can be metastably retained at ambient pressure. Deng and Chu [10] demonstrated that pressure-quenched HgBaCaCuO retains $T_c = 151$ K at ambient pressure, proving the viability of the approach for complex oxides.

Proposed protocol.—(i) Synthesize LaH_{10} at 170 GPa, 2000 K in a diamond anvil cell (DAC) [28]. (ii) Cool to 4 K under full pressure. (iii) Rapidly decompress ($dP/dt > 10$ GPa/s) while maintaining $T < 20$ K. (iv) Warm slowly, measuring $R(T)$.

The rapid low-temperature quench exploits kinetic trapping: at 4 K, hydrogen diffusion is negligible ($D_H \sim 10^{-30}$ cm²/s), so the clathrate cage cannot decompose on experimental timescales. Even partial retention of the structure (R_τ reduced from 4.24 to ~ 1.5) would yield $T_c \sim 180$ –200 K at ambient pressure—near 0°C .

Risk: LaH_{10} may decompose via hydrogen escape through grain boundaries even at low T . Mitigation: encapsulate in an amorphous boron nitride shell before decompression.

See Supplemental Material [1], Sec. S7B for the complete DAC protocol.

C. Prediction 3: Multi-channel enhancement in $\text{La}_3\text{Ni}_2\text{O}_7 + \text{H}$

$\text{La}_3\text{Ni}_2\text{O}_7$ already achieves $R_\tau^{\text{multi}} = 1.47$ at 14 GPa through 3 pairing channels (phonon + spin + Ni 3d orbital). Adding hydrogen via topotactic hydrogenation would introduce a fourth channel: hydrogen vibrational modes coupling to the Ni 3d electrons.

The expected T_c enhancement is

$$\frac{T_c^{\text{H}}}{T_c^{(0)}} \approx \frac{R_\tau^{\text{multi}}(4 \text{ ch})}{R_\tau^{\text{multi}}(3 \text{ ch})} = \frac{4}{3}, \quad (28)$$

predicting $T_c^{\text{H}} \approx 107 \pm 25$ K at 14 GPa, or $T_c \approx 84 \pm 20$ K for the ambient-pressure film.

D. Prediction 4: $\text{Eu}_2\text{H}_6\text{Ir}$ (4-channel candidate)

Among the 897 screened compounds, $\text{Eu}_2\text{H}_6\text{Ir}$ has the highest $R_\tau^{\text{multi}} = 0.38$ at ambient pressure, with four identified channels: (i) phonon (H vibrational modes, $\omega_H \approx 100$ meV); (ii) spin (Eu 4f⁷, $S = 7/2$, large local moment); (iii) topological (Ir 5d spin-orbit coupling, potentially nontrivial $\mathbb{Z}_2$ invariant); (iv) orbital (Ir 5d–Eu 5d hybridization).

Despite $R_\tau^{\text{multi}} < 1$, this compound may exhibit superconductivity at $T_c \approx 10\text{--}30\text{ K}$ via the multi-channel mechanism. It has not been synthesized.

Proposed synthesis.—Arc-melt Eu + Ir in a 1:0.5 stoichiometric ratio under argon. Hydrogenate the resulting Eu_2Ir ingot at 300 K, 50 bar H_2 for 24 h. Characterize structure by XRD; measure $R(T)$ down to 2 K.

VIII. CONNECTION TO QUANTUM PROCESSOR EXPERIMENTS

The τ framework has a direct experimental antecedent. On the QuTech Tuna-9 superconducting processor [14], a single Bell pair reduced τ from 0.47 to 0.05 (89% reduction). This is the same physics as Cooper pairing:

	Bell pair (Tuna-9)	Cooper pair
Entanglement	$\ln 2$ nats	$\ln 2$ nats (at $\mathbf{k}_F$)
Bare τ	0.47	$\tau_{\text{ph}}(T)$
Assisted τ	0.05	0 (below T_c)
Recovery efficiency	89%	100% (macroscopic)

The Cooper pair achieves 100% recovery (vs. 89% for Tuna-9) because the BCS condensate involves $\sim 10^{23}$ entangled pairs acting collectively, whereas Tuna-9 used a single pair on a noisy channel. Future quantum processor experiments with tunable noise could directly simulate the R_τ criterion for candidate materials [14].

IX. DISCUSSION

A. What the R_τ criterion adds

The R_τ screening criterion provides three capabilities absent from existing approaches:

(i) *Mechanism-independent comparison.*—Table II places conventional (Nb), two-gap (MgB_2), cuprate (YBCO), hydride (H_3S), and nickelate ($\text{La}_3\text{Ni}_2\text{O}_7$) superconductors on a single quantitative scale. This is impossible with the McMillan formula alone, which requires knowledge of the pairing mechanism.

(ii) *High-throughput screening.*—The proxy $R_\tau \approx 1.22 T_c \Theta_D / T^2$ requires only two numbers per material and can be evaluated for entire databases. Our screening of 897 compounds took < 1 CPU-hour.

(iii) *Design guidance.*—The multi-channel criterion (26) quantifies the advantage of combining pairing mechanisms, providing a concrete target for materials design.

B. Relation to prior work on entanglement in superconductors

Previous work has established that the BCS ground state contains macroscopic entanglement [44, 53], that Cooper pairs can be viewed as entangled pairs [52, 54], and that information-theoretic measures detect the superconducting transition [51]. Amico et al. [50] provide a comprehensive review of entanglement in many-body systems, and Brandão [55] analyzed entanglement quantification in mixed states relevant to finite-temperature superconductors. Most recently, Huo et al. [56] applied the Petz recovery map to diagnose quantum phases in topological systems.

Our contribution is distinct: we use the Petz recovery framework not as a diagnostic but as a *mechanism*—proposing that the entanglement of Cooper pairs directly cancels the temporal asymmetry of electron transport, yielding the R_τ screening criterion. This mechanistic role of entanglement is what produces the quantitative material predictions absent in prior information-theoretic analyses of superconductivity.

C. Why room temperature is hard

The screening result—no ambient-pressure single-channel material among 897 hydrogen-metallic compounds reaches $R_\tau = 1$ —has a simple physical explanation. At $T = 300\text{ K}$, the phonon decoherence $\tau_{\text{ph}} \approx \pi\lambda \cdot 300 / \Theta_D$ grows linearly with T . To compensate, $I_{\text{Cooper}} \propto \Delta(0) \propto \Theta_D e^{-1/\lambda}$ must be exponentially large in $1/\lambda$. The product $T_c \Theta_D$ is bounded from above [27]: for Eliashberg phonon-mediated superconductors at ambient pressure, $T_c \Theta_D \lesssim 30,000\text{ K}^2$, which falls short of the single-channel threshold $73,800\text{ K}^2$ by a factor of ~ 2.5 .

The high-pressure hydrides circumvent this by achieving $\Theta_D > 1000\text{ K}$ through the light hydrogen mass and stiff covalent bonds. The challenge of room-temperature superconductivity at ambient pressure thus reduces to: *how to retain $\Theta_D > 1000\text{ K}$ without megabar pressures?* The R_τ framework suggests three routes: metastable quenching, multi-channel pairing, or a new mechanism.

D. Limitations

(i) The proxy (22) assumes weak coupling and neglects retardation effects. For strong-coupling materials ($\lambda > 2$), full Eliashberg calculations are needed to validate R_τ .

(ii) The multi-channel enhancement assumes independent (incoherent) addition. Inter-channel coherence may enhance or suppress R_τ^{multi} ; the conditions for constructive interference require symmetry analysis on a case-by-case basis.

(iii) The Materials Project screening relies on estimated λ values with $\sim 30\text{--}50\%$ uncertainty.

First-principles electron-phonon calculations (e.g., via EPW [34]) would sharpen the predictions but are computationally expensive for all 897 compounds.

(iv) The topological contribution $I_{\text{topo}} = |\nu| \ln 2$ is temperature-independent by construction, but its magnitude for specific materials requires knowledge of the band topology, which is not available in the Materials Project for most entries.

X. CONCLUSION

We have developed the R_τ screening criterion for superconductivity from the quantum information framework $\tau = 1 - F$. The criterion is mechanism-independent, computationally inexpensive, and recovers the BCS gap equation as a special case. Screening 897 hydrogen-metallic compounds reveals that ambient-pressure room-temperature superconductivity via single-channel phonon pairing is extremely unlikely, but multi-channel materials and pressure-quenched hydrides remain viable.

While pressure quenching of known hydrides (LaH₁₀, H₃S) provides the most direct route to near-0 °C superconductivity, the long-term goal of ambient-pressure room-temperature superconductivity requires discovering new materials. The R_τ framework provides three concrete strategies: (1) light-element scaffolds for maximum

Θ_D , (2) multi-channel s -wave pairing, and (3) metastable phase engineering. We have identified three specific material families (clathrate hydrides, nickelate hydrides, diamond-scaffold hydrides) as priority targets for experimental synthesis and first-principles screening.

Our most testable suggestion is that LaNi₅H₆—a ubiquitous hydrogen storage alloy—may superconduct at $T_c \approx 31 \pm 12$ K based on heuristic coupling estimates. This can be verified with standard cryogenic equipment in any condensed matter laboratory. If confirmed, it would validate the R_τ framework and open a systematic search for higher- T_c multi-channel materials.

The deepest lesson is that a Cooper pair is a quantum error correction code: the BCS condensate is a macroscopic ensemble of entanglement-assisted recovery channels, each protecting electronic quantum information against thermal decoherence [13, 14]. Room-temperature superconductivity requires codes that can correct errors at a rate set by 300 K phonons—a formidable but perhaps not impossible challenge.

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- [1] See Supplemental Material at [URL] for detailed derivations, material-by-material calculations, experimental protocols, and the R_τ calculator code.
 - [2] H. K. Onnes, The resistance of pure mercury at helium temperatures, *Commun. Phys. Lab. Univ. Leiden* **12**, 120 (1911).
 - [3] J. Bardeen, L. N. Cooper, and J. R. Schrieffer, Theory of superconductivity, *Phys. Rev.* **108**, 1175 (1957).
 - [4] J. G. Bednorz and K. A. Müller, Possible high T_c superconductivity in the Ba–La–Cu–O system, *Z. Phys. B* **64**, 189 (1986).
 - [5] A. Schilling, M. Cantoni, J. D. Guo, and H. R. Ott, Superconductivity above 130 K in the Hg–Ba–Ca–Cu–O system, *Nature* **363**, 56 (1993).
 - [6] A. P. Drozdov, M. I. Erements, I. A. Troyan, V. Ksenofontov, and S. I. Shylin, Conventional superconductivity at 203 kelvin at high pressures in the sulfur hydride system, *Nature* **525**, 73 (2015).
 - [7] M. Somayazulu *et al.*, Evidence for superconductivity above 260 K in lanthanum superhydride at megabar pressures, *Phys. Rev. Lett.* **122**, 027001 (2019).
 - [8] H. Sun *et al.*, Signatures of superconductivity near 80 K in a nickelate under high pressure, *Nature* **621**, 493 (2023).
 - [9] E. Ko *et al.*, Signatures of ambient pressure superconductivity in thin film La₃Ni₂O₇, *Nature* **638**, 935 (2025).
 - [10] D. Deng and C. W. Chu, Pressure-quenched HgBaCaCuO with $T_c = 151$ K at ambient pressure, Preprint (2026).
 - [11] W. L. McMillan, Transition temperature of strong-coupled superconductors, *Phys. Rev.* **167**, 331 (1968).
 - [12] P. B. Allen and R. C. Dynes, Transition temperature of strong-coupled superconductors reanalyzed, *Phys. Rev. B* **12**, 905 (1975).
 - [13] S.-K. Huang, Temporal asymmetry as Petz recovery failure, Zenodo DOI: 10.5281/zenodo.18897853 (2026); see also <https://github.com/akaiHuang/petz-recovery-unification>.
 - [14] S.-K. Huang, Measuring temporal asymmetry on a superconducting quantum processor: From quantum erasure to the birth of time’s arrow, Preprint (2026).
 - [15] A. Jain *et al.*, Commentary: The Materials Project: A materials genome approach to accelerating materials innovation, *APL Mater.* **1**, 011002 (2013).
 - [16] C. Chen *et al.*, A critical review of machine learning of energy materials, *Adv. Energy Mater.* **10**, 1903242 (2020).
 - [17] V. Stanev *et al.*, Machine learning modeling of superconducting critical temperature, *npj Comput. Mater.* **4**, 29 (2018).
 - [18] D. Petz, Sufficiency of channels over von Neumann algebras, *Q. J. Math.* **39**, 97 (1988).
 - [19] A. Uhlmann, The “transition probability” in the state space of a $*$ -algebra, *Rep. Math. Phys.* **9**, 273 (1976).
 - [20] M. Junge, R. Renner, D. Sutter, M. M. Wilde, and A. Winter, Universal recovery maps and approximate sufficiency of quantum relative entropy, *Ann. Henri Poincaré* **19**, 2955 (2018).
 - [21] O. Fawzi and R. Renner, Quantum conditional mutual

- information and approximate Markov chains, Commun. Math. Phys. **340**, 575 (2015).
- [22] I. Devetak and A. Winter, Distillation of secret key and entanglement from quantum states, Proc. R. Soc. A **461**, 207 (2005).
- [23] G. M. Eliashberg, Interactions between electrons and lattice vibrations in a superconductor, Sov. Phys. JETP **11**, 696 (1960).
- [24] H. J. Choi, D. Roundy, H. Sun, M. L. Cohen, and S. G. Louie, The origin of the anomalous superconducting properties of MgB_2 , Nature **418**, 758 (2002).
- [25] H. Liu, I. I. Naumov, R. Hoffmann, N. W. Ashcroft, and R. J. Hemley, Potential high- T_c superconducting lanthanum and yttrium hydrides at high pressure, Proc. Natl. Acad. Sci. USA **114**, 6990 (2017).
- [26] I. Errea *et al.*, Quantum crystal structure in the 250-kelvin superconducting lanthanum hydride, Nature **578**, 66 (2020).
- [27] M. Gao *et al.*, Upper bound on the critical temperature of superconductors from first-principles phonon calculations, Phys. Rev. B **111**, 024506 (2025).
- [28] A. P. Drozdov *et al.*, Superconductivity at 250 K in lanthanum hydride under high pressures, Nature **569**, 528 (2019).
- [29] J. H. N. van Vucht, F. A. Kuijpers, and H. C. A. M. Bruning, Reversible room-temperature absorption of large quantities of hydrogen by intermetallic compounds, Philips Res. Rep. **25**, 133 (1970).
- [30] A. Percheron-Guégan, C. Lartigue, J. C. Achard, P. Germi, and F. Tasset, Neutron and X-ray diffraction profile analyses and structure of LaNi_5 , $\text{LaNi}_{5-x}\text{Al}_x$ and $\text{LaNi}_{5-x}\text{Mn}_x$ intermetallics and their hydrides, J. Less-Common Met. **74**, 1 (1980).
- [31] M. Gupta, Electronic structure and electron-phonon coupling in LaNi_5 , J. Less-Common Met. **130**, 219 (1987).
- [32] D. Richter, R. Hempelmann, and R. C. Bowman, Dynamics of hydrogen in intermetallic hydrides, in *Hydrogen in Intermetallic Compounds II*, edited by L. Schlapbach (Springer, Berlin, 1992), p. 97.
- [33] V. Paul-Boncour *et al.*, Structural and magnetic properties of ErFe_2H_x and YFe_2H_x compounds ($0 \leq x \leq 5$), J. Solid State Chem. **142**, 120 (1999).
- [34] S. Poncé, E. R. Margine, C. Verdi, and F. Giustino, EPW: Electron-phonon coupling, transport and superconducting properties using maximally localized Wannier functions, Comput. Phys. Commun. **209**, 116 (2016).
- [35] P. W. Anderson, The resonating valence bond state in La_2CuO_4 and superconductivity, Science **235**, 1196 (1987).
- [36] D. J. Scalapino, E. Loh, and J. E. Hirsch, d -wave pairing near a spin-density-wave instability, Phys. Rev. B **34**, 8190 (1986).
- [37] T. Moriya and K. Ueda, Spin fluctuations and high temperature superconductivity, Adv. Phys. **49**, 555 (2000).
- [38] H. Kontani and S. Onari, Orbital-fluctuation-mediated superconductivity in iron pnictides: Analysis of the five-orbital Hubbard–Holstein model, Phys. Rev. Lett. **104**, 157001 (2010).
- [39] L. Fu and C. L. Kane, Superconducting proximity effect and Majorana fermions at the surface of a topological insulator, Phys. Rev. Lett. **100**, 096407 (2008).
- [40] M. Sato and Y. Ando, Topological superconductors: A review, Rep. Prog. Phys. **80**, 076501 (2017).
- [41] J. M. Ziman, *Electrons and Phonons* (Oxford University Press, Oxford, 1960).
- [42] D. Reeb and M. M. Wolf, An improved Landauer principle with finite-size corrections, New J. Phys. **16**, 103011 (2014).
- [43] M. Tinkham, *Introduction to Superconductivity*, 2nd ed. (Dover, New York, 2004).
- [44] V. Vedral, High-temperature macroscopic entanglement, New J. Phys. **6**, 102 (2004).
- [45] E. Zurek and T. Bi, High-temperature superconductivity in alkaline and rare earth polyhydrides at high pressure: A theoretical perspective, J. Chem. Phys. **150**, 050901 (2019).
- [46] D. C. Mattis and J. Bardeen, Theory of the anomalous skin effect in normal and superconducting metals, Phys. Rev. **111**, 412 (1958).
- [47] L. Boeri *et al.*, The 2021 room-temperature superconductivity roadmap, J. Phys. Condens. Matter **34**, 183002 (2022).
- [48] J. A. Flores-Livas *et al.*, A perspective on conventional high-temperature superconductors at high pressure: Methods and materials, Phys. Rep. **856**, 1 (2020).
- [49] C. J. Pickard, I. Errea, and M. I. Erements, Superconducting hydrides under pressure, Annu. Rev. Condens. Matter Phys. **11**, 57 (2020).
- [50] L. Amico, R. Fazio, A. Osterloh, and V. Vedral, Entanglement in many-body systems, Rev. Mod. Phys. **80**, 517 (2008).
- [51] W. Ding *et al.*, Diagnosing quantum phase transitions via holographic entanglement entropy, Proc. Natl. Acad. Sci. USA **118**, e2104114118 (2021).
- [52] S. Oh and J. Kim, Entanglement of electron spins of noninteracting electron gases, arXiv:quant-ph/0406013 (2004).
- [53] C. Dunning, J. Links, and H.-Q. Zhou, Ground-state entanglement of the BCS model, Phys. Rev. Lett. **94**, 227002 (2005).
- [54] A. Botero, Quantum information and the BCS ground state, arXiv:quant-ph/0404176 (2004).
- [55] F. G. S. L. Brandão, Quantifying entanglement with witness operators, Phys. Rev. A **72**, 022310 (2005).
- [56] M. Huo *et al.*, Petz recovery map diagnostics of quantum phases, Phys. Rev. B **110**, 195107 (2024).